

Euclidean geometry – 1

Vectors, angles, simplices and volumes in $\mathbb{R}^n$

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Theoretical minimum. The whole metric geometry of finitely many points of $\mathbb{R}^n$ is stored in the Gram matrix of their difference vectors: lengths, angles, volumes and centres are read off from it, and a question of the type “how many vectors with prescribed angles fit into $\mathbb{R}^n$ ” becomes a question about the rank of a positive semidefinite matrix.

Definitions. The *Gram matrix* of $v_1, \dots, v_k \in \mathbb{R}^n$ is $\Gamma(v_1, \dots, v_k) = (\langle v_i, v_j \rangle) = A^T A$, where $A = (v_1 | \dots | v_k)$; it is positive semidefinite and its rank is the rank of the system (Linear algebra – 5). An *n-simplex* $S = A_0 A_1 \dots A_n$ is the convex hull of $n + 1$ points not lying in one hyperplane, and $v(S)$ is its volume. Every $X \in \mathbb{R}^n$ is uniquely $X = \sum \lambda_i A_i$ with $\sum \lambda_i = 1$ (*barycentric coordinates*); $|\lambda_i| = v(S_i)/v(S)$, where S_i is S with A_i replaced by X , and $\lambda_i > 0$ for all i exactly when X is interior. The *centroid* is $G = \frac{1}{n+1} \sum A_i$, the *circumcentre* O is the centre of the sphere through all vertices. Points are in *general position* if no $n + 1$ of them lie in one hyperplane.

Theorem 1 (Gram and Cayley–Menger). The k -dimensional volume of the parallelepiped spanned by $v_1, \dots, v_k$ equals $\sqrt{\det \Gamma(v_1, \dots, v_k)}$; hence $v(S) = \frac{1}{n!} \sqrt{\det \Gamma(A_1 - A_0, \dots, A_n - A_0)}$. Since $2\langle A_i - A_0, A_j - A_0 \rangle = d_{0i}^2 + d_{0j}^2 - d_{ij}^2$, where $d_{ij} = |A_i A_j|$, the volume depends only on the distances:

$$(-1)^{n+1} 2^n (n!)^2 v(S)^2 = \det \begin{pmatrix} 0 & \mathbf{1}^T \\ \mathbf{1} & (d_{ij}^2)_{i,j=0}^n \end{pmatrix}.$$

Proof idea. For $k = n$, $\det \Gamma = \det(A^T A) = (\det A)^2$; in general apply this inside the span of the v_i . The bordered form follows by row operations (Linear algebra – 8, homework 8).

Theorem 2 (Pythagoras for volumes). For a k -dimensional parallelepiped P in $\mathbb{R}^n$ and the orthogonal projections π_I onto the coordinate k -planes, $\text{vol}_k(P)^2 = \sum_{|I|=k} \text{vol}_k(\pi_I P)^2$: this is $\det(A^T A) = \sum_I (\det A_I)^2$,

i.e. Cauchy–Binet (Linear algebra – 8). A projection restricted to a k -plane is linear, so the identity holds for every measurable subset of a k -plane; for $k = n - 1$ the projection onto $u^\perp$ multiplies areas in a hyperplane with unit normal ν by $|\langle \nu, u \rangle|$.

Theorem 3 (non-acute and obtuse systems). Let $v_1, \dots, v_m \in \mathbb{R}^n$ be non-zero. If $\langle v_i, v_j \rangle \leq 0$ for all $i \neq j$, then $m \leq 2n$; if $\langle v_i, v_j \rangle < 0$ for all $i \neq j$, then $m \leq n + 1$. *Proof sketch.* The projections w_i of v_i onto $v_m^\perp$ satisfy $\langle w_i, w_j \rangle = \langle v_i, v_j \rangle - \langle v_i, v_m \rangle \langle v_j, v_m \rangle / |v_m|^2 \leq \langle v_i, v_j \rangle$, and at most one of them vanishes (none in the obtuse case, $m \geq 3$); induct on n . A useful lemma: if $w = \sum_{i \in I} \lambda_i v_i = \sum_{j \in J} \mu_j v_j$ with disjoint I, J and positive coefficients, then $|w|^2 = \sum \lambda_i \mu_j \langle v_i, v_j \rangle \leq 0$, so $w = 0$.

Theorem 4 (equiangular lines, Gerzon). At most $\frac{n(n+1)}{2}$ lines through the origin of $\mathbb{R}^n$ make pairwise equal angles. *Proof.* For unit directions with $|\langle v_i, v_j \rangle| = \alpha < 1$ ($i \neq j$) the matrices $P_i = v_i v_i^T$ have the Gram matrix $(\text{tr } P_i P_j) = (1 - \alpha^2)E + \alpha^2 J \succ 0$, so they are linearly independent in the $\frac{n(n+1)}{2}$ -dimensional space of symmetric matrices.

Fact 1 (centres). $\sum_i |XA_i|^2 = (n + 1)|XG|^2 + \sum_i |GA_i|^2$ for every X (Leibniz); the circumcentre solves the linear system $2\langle O - A_0, A_i - A_0 \rangle = |A_i - A_0|^2$, $1 \leq i \leq n$, with the matrix $2\Gamma(A_1 - A_0, \dots, A_n - A_0)$. Plane combinatorial geometry is Combinatorics – 6, and Helly, Radon and Carathéodory are Convex – 1.

1. Prove Theorem 3 and show that both bounds are attained (compare Methods of proof – 1, homework 6). Then prove that $2n$ non-zero vectors in $\mathbb{R}^n$ with pairwise non-acute angles always lie on n pairwise orthogonal lines, two opposite vectors on each line; and that unit vectors $v_1, \dots, v_m$ with $\langle v_i, v_j \rangle \leq -\beta$ for all $i \neq j$, where $\beta > 0$, satisfy $m \leq 1 + 1/\beta$.
2. Let F_i denote the $(n - 1)$ -dimensional volume of the facet opposite A_i of an n -simplex $A_0 A_1 \dots A_n$. (a) Prove that if the edges $A_0 A_1, \dots, A_0 A_n$ are pairwise orthogonal, then $F_0^2 = F_1^2 + \dots + F_n^2$ (for $n = 3$ this is de Gua's theorem). (b) Prove that $\sum_{i=0}^n F_i u_i = 0$ for every n -simplex, where u_i is the outer unit normal of the facet opposite A_i . Deduce that

$F_0 < F_1 + \dots + F_n$ and that $F_0^2 = \sum_{i=1}^n F_i^2 - 2 \sum_{1 \leq i < j \leq n} F_i F_j \cos \varphi_{ij}$, where φ_{ij} is the dihedral angle between the facets opposite A_i and A_j .

3. Let $n \geq 2$, let $A_1, A_2, \dots, A_{n+1}$ be $n+1$ points in the n -dimensional Euclidean space, not lying on the same hyperplane, and let B be a point strictly inside the convex hull of $A_1, A_2, \dots, A_{n+1}$. Prove that $\angle A_i B A_j > 90^\circ$ holds for at least n pairs (i, j) with $1 \leq i < j \leq n+1$.

(IMC 2015)

4. Given an n -simplex with vertices $A_1, A_2, \dots, A_{n+1}$ ($n \geq 2$) and any point M in its interior, let N_i be the point on the hyperplane opposite A_i such that A_i, M and N_i are collinear. Find the point M for which

$$\frac{\text{Volume}(N_1 N_2 \dots N_{n+1})}{\text{Volume}(A_1 A_2 \dots A_{n+1})}$$

is maximal and find this maximum.

(IMC 2025, shortlist)

5. Let n be a positive integer. An n -simplex in $\mathbb{R}^n$ is given by $n+1$ points $P_0, P_1, \dots, P_n$, called its vertices, which do not all belong to the same hyperplane. For every n -simplex S denote by $v(S)$ its volume and by $C(S)$ the centre of the unique sphere containing all the vertices of S . Suppose that P is a point inside an n -simplex S , and let S_i be the n -simplex obtained from S by replacing its i -th vertex by P . Prove that

$$v(S_0)C(S_0) + v(S_1)C(S_1) + \dots + v(S_n)C(S_n) = v(S)C(S).$$

(IMC 2009)